When X is 20, Y will be

$Y = -0.25\,(20)+44.25$

$= -5+44.25$

$= 39.25$ (when the price is ₹20, the likely demand is 39.25)

Example 9.11

Obtain regression equation of Y on X and estimate Y when $X = 55$ from the following:

X	40	50	38	60	65	50	35
Y	38	60	55	70	60	48	30

Solution:

X	Y	$dx = (X-48)$	dx^2	$dy = (Y-50)$	dy^2	$dxdy$
40	38	–8	64	–12	144	96
50	60	2	4	10	100	20
38	55	–10	100	5	25	–50
60	70	12	144	20	400	240
65	60	17	289	10	100	170
50	48	2	4	-2	4	-4
35	30	-13	169	-20	400	260
$\Sigma X =$ 338	$\Sigma Y =$ 361	$\Sigma dx =$ 2	$\Sigma dx^2 =$ 774	$\Sigma dy =$ 11	$\Sigma dy^2 =$ 1173	$\Sigma dxy =$ 732

Table 9.9

$$\overline{X} = \frac{\Sigma X}{N} = \frac{338}{7} = 48.29$$

$$\overline{Y} = \frac{\Sigma Y}{N} = \frac{361}{7} = 51.57$$

(i) Regression coefficients of Y on X:

$$b_{yx} = \frac{N\Sigma dx\,dy - (\Sigma dx)(\Sigma dy)}{N\,\Sigma dx^2 - (\Sigma dx)^2}$$

$$= \frac{7(732)-(2)(11)}{7(774)-(2)^2}$$

$$= \frac{5124-22}{5418-4}$$

$$= \frac{5102}{5414}$$

$= 0.942$

$b_{yx} = 0.942$

(ii) Regression equation of Y on X:

$Y - \overline{Y} = b_{yx}(X - \overline{X})$

$Y-51.57 = 0.942(X-48.29)$

$Y = 0.942X-45.49+51.57=0.942\times -45.49+51.57$

$Y = 0.942X+6.08$

$\therefore$ The regression equation of Y on X is $Y= 0.942X+6.08$

Estimation of Y when $X= 55$

$Y= 0.942(55)+6.08=57.89$

Example 9.12

Find the means of X and Y variables and the coefficient of correlation between them from the following two regression equations:

$2Y-X-50 = 0$

$3Y-2X-10 = 0.$

Solution:

We are given

$2Y-X-50 = 0$ ... (1)

$3Y-2X-10 = 0$ ... (2)

Solving equation (1) and (2)

We get $Y= 90$

Putting the value of Y in equation (1)

We get $X = 130$

Hence $\overline{X} = 130$ and $\overline{Y} = 90$

Calculating correlation coefficient

Let us assume equation (1) be the regression equation of Y on X.

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